AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 10

Lab Title Graphical User Interface

Student Name: Ali Abbas Khan

Reg. No: 230661

Objective: _____

_____

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 10

Title: GUI

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date**:** December 3, 2025

Lab Task:

1. **Design a simple calculator using MATLAB GUI that can add, subtract, multiply and divide two numbers and displays the result. Make Sure this should be colourful and presentable.**

Code:

C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L10\Task1.m

EDITOR | PUBLISH | VIEW

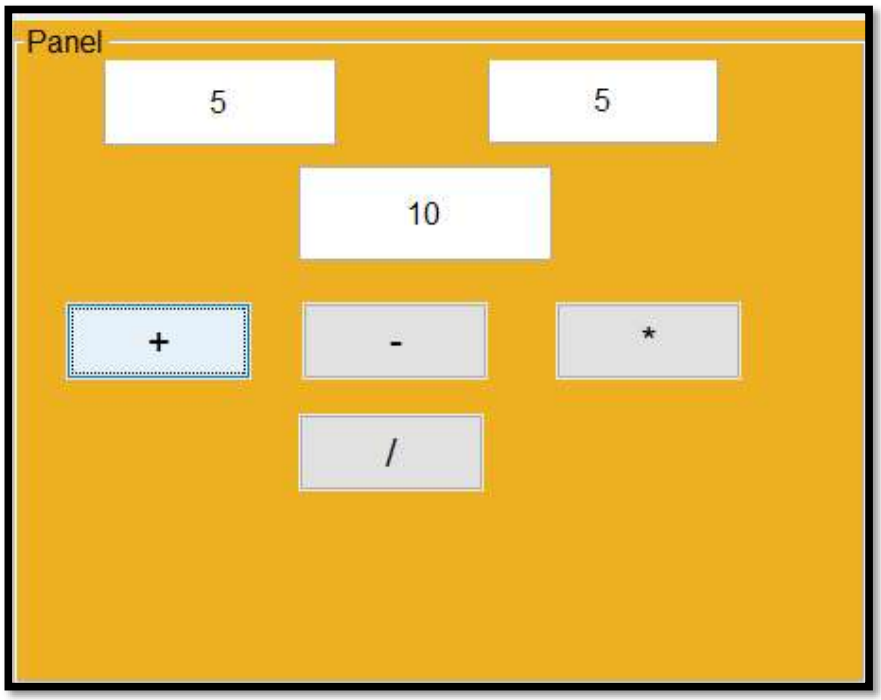
New | Open | Save | Find Files | Compare | Print | Go To | Find | Insert | Comment | Indent | Breakpoints | Run | Run and Advance | Run Section | Advance | Run and Time

FILE | NAVIGATE | EDIT | BREAKPOINTS | RUN

```
146 function pushbutton1_Callback(hObject, eventdata, handles)
147 % hObject    handle to pushbutton1 (see GCBO)
148 % eventdata  reserved - to be defined in a future version of MATLAB
149 % handles    structure with handles and user data (see GUIDATA)
150 num1=str2num(get(handles.edit1,'string'));
151 num2=str2num(get(handles.edit4,'string'));
152 result=num1+num2;
153 set(handles.edit3,'string',num2str(result));
154
155
156 % --- Executes on button press in pushbutton2.
157 function pushbutton2_Callback(hObject, eventdata, handles)
158 % hObject    handle to pushbutton2 (see GCBO)
159 % eventdata  reserved - to be defined in a future version of MATLAB
160 % handles    structure with handles and user data (see GUIDATA)
161 num1=str2num(get(handles.edit1,'string'));
162 num2=str2num(get(handles.edit4,'string'));
163 result=num1-num2;
164 set(handles.edit3,'string',num2str(result));
165
166 % --- Executes on button press in pushbutton3.
167 function pushbutton3_Callback(hObject, eventdata, handles)
168 % hObject    handle to pushbutton3 (see GCBO)
169 % eventdata  reserved - to be defined in a future version of MATLAB
170 % handles    structure with handles and user data (see GUIDATA)
171 num1=str2num(get(handles.edit1,'string'));
172 num2=str2num(get(handles.edit4,'string'));
173 result=num1*num2;
174 set(handles.edit3,'string',num2str(result));
```

```
% --- Executes on button press in pushbutton4.
function pushbutton4_Callback(hObject, eventdata, handles)
% hObject    handle to pushbutton4 (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles    structure with handles and user data (see GUIDATA)
num1=str2num(get(handles.edit1,'string'));
num2=str2num(get(handles.edit4,'string'));
if num2~=0
    result=num1/num2;
else
    result = NaN;
end
set(handles.edit3,'string',num2str(result));
```

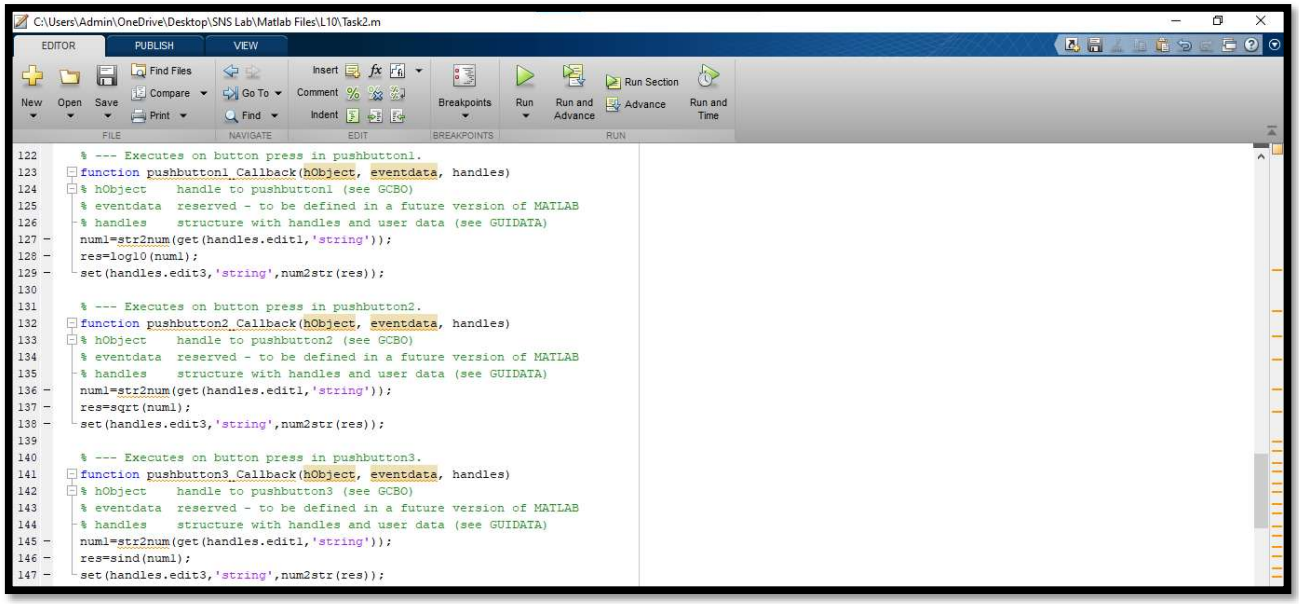
Output:

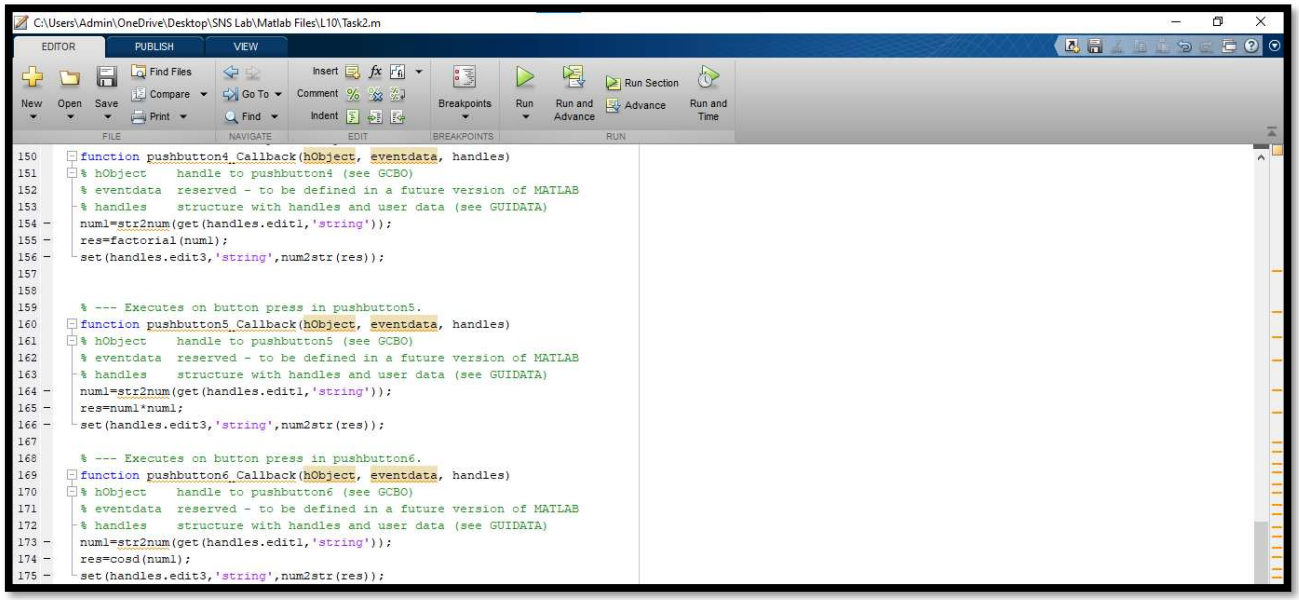


Design a trigonometric calculator using MATLAB GUI that can take

- Natural log
- Factorial
- log10
- exp
- square root
- square
- sin
- cos
- tan
- of any input number

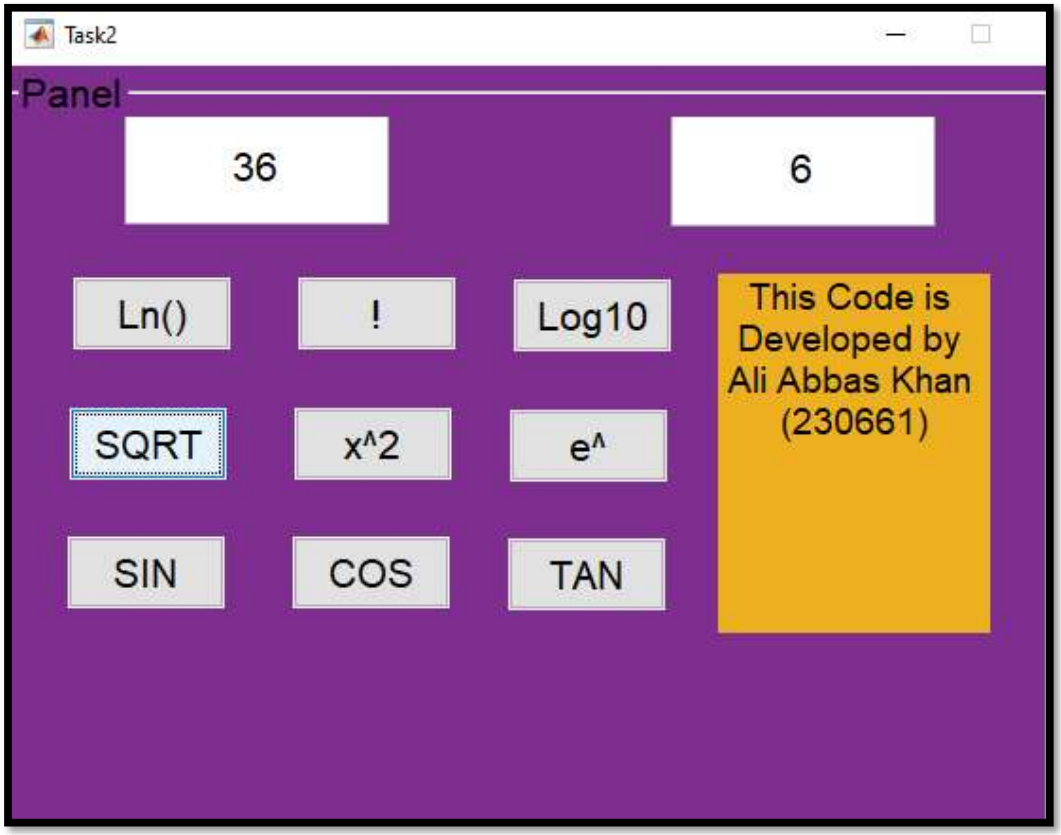
Code:





```
177    % --- Executes on button press in pushbutton7.
178    function pushbutton7_Callback(hObject, eventdata, handles)
179    % hObject    handle to pushbutton7 (see GCBO)
180    % eventdata  reserved - to be defined in a future version of MATLAB
181    % handles    structure with handles and user data (see GUIDATA)
182 -  num1=str2num(get(handles.edit1,'string'));
183 -  res=log10(num1);
184 -  set(handles.edit3,'string',num2str(res));
185
186
187    % --- Executes on button press in pushbutton8.
188    function pushbutton8_Callback(hObject, eventdata, handles)
189    % hObject    handle to pushbutton8 (see GCBO)
190    % eventdata  reserved - to be defined in a future version of MATLAB
191    % handles    structure with handles and user data (see GUIDATA)
192 -  num1=str2num(get(handles.edit1,'string'));
193 -  res=exp(num1);
194 -  set(handles.edit3,'string',num2str(res));
195
196    % --- Executes on button press in pushbutton9.
197    function pushbutton9_Callback(hObject, eventdata, handles)
198    % hObject    handle to pushbutton9 (see GCBO)
199    % eventdata  reserved - to be defined in a future version of MATLAB
200    % handles    structure with handles and user data (see GUIDATA)
201 -  num1=str2num(get(handles.edit1,'string'));
202 -  res=tand(num1);
203 -  set(handles.edit3,'string',num2str(res));
```

Output:



Conclusion:

The development of the MATLAB GUI demonstrates how graphical interfaces can simplify complex tasks by providing an interactive and user-friendly environment. Through this project, the integration of code with visual elements allowed efficient data handling, clear visualization, and improved usability compared to command-line execution. Overall, the GUI highlights the practical benefits of combining MATLAB's computational power with intuitive design, making the application more accessible and effective for end users.